



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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ASSIGNMENT No: 4

TITLE: Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory

Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.



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Exercise

1. Create a program to convert student marks from String format to Integer and calculate total marks.

Input and Output combined-

```
1 public class StudentMarks {
2
3     public static void main(String[] args) {
4
5         // Marks stored as Strings
6         String mark1 = "85";
7         String mark2 = "90";
8         String mark3 = "78";
9
10        // Convert String to Integer
11        int m1 = Integer.parseInt(mark1);
12        int m2 = Integer.parseInt(mark2);
13        int m3 = Integer.parseInt(mark3);
14
15        // Calculate total marks
16        int total = m1 + m2 + m3;
17
18        System.out.println("Marks in Subject 1 = " + m1);
19        System.out.println("Marks in Subject 2 = " + m2);
20        System.out.println("Marks in Subject 3 = " + m3);
21        System.out.println("Total Marks = " + total);
22    }
23 }
24
```

Console Output:

```
<terminated> StudentMarks [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (30-Jul-2026, 11:29:52 am - 11:29:52 am elapsed 0:00:0)
Marks in Subject 1 = 85
Marks in Subject 2 = 90
Marks in Subject 3 = 78
Total Marks = 253
```



2. Develop an Employee Payroll System that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

Input-

```
1 import java.util.Scanner;
2 public class EmployeePayroll {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter Employee ID: ");
6         String empId = sc.nextLine();
7         System.out.print("Enter Basic Salary: ");
8         String basicSalary = sc.nextLine();
9         System.out.print("Enter Bonus Amount: ");
10        String bonus = sc.nextLine();
11        // Convert to Wrapper Objects
12        Integer salary = Integer.valueOf(basicSalary);
13        Integer bonusAmount = Integer.valueOf(bonus);
14        // Validation
15        if (salary < 0 || bonusAmount < 0) {
16            System.out.println("Invalid salary or bonus amount.");
17        } else {
18            int netSalary = salary + bonusAmount;
19            System.out.println("\nEmployee ID : " + empId);
20            System.out.println("Basic Salary : " + salary);
21            System.out.println("Bonus Amount : " + bonusAmount);
22            System.out.println("Net Salary : " + netSalary);
23        }
24        sc.close();
25    }
26 }
```

Output-

```
Enter Employee ID: 101
Enter Basic Salary: 1000000
Enter Bonus Amount: 10
```

```
Employee ID : 101
Basic Salary : 1000000
Bonus Amount : 10
Net Salary : 1000010
```